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Influence of total-body mass on the scaling of *S*-factors for patient-specific, blood-based red-marrow dosimetry

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Abstract

To perform patient-specific, blood-based red-marrow dosimetry, dose conversion factors (the *S* factors in the MIRD formalism) have to be scaled by patients' organ masses. The dose to red marrow includes both self-dose and cross-irradiation contributions. Linear mass scaling for the self-irradiation term only is usually applied as a first approximation, whereas the cross-irradiation term is considered to be mass independent. Recently, the need of a mass scaling correction on both terms, not necessarily linear and dependent on the radionuclide, has been highlighted in the literature. *S*-factors taking into account different mass adjustments of organs are available in the OLINDA/EXM code. In this paper, a general algorithm able to fit the mass-dependent factors $S_{\mathrm{rm}\leftarrow\mathrm{tb}}$ and $S_{\mathrm{rm}\leftarrow\mathrm{rm}}$ is suggested and included in a more general equation for red-marrow dose calculation. Moreover, parameters to be considered specifically for therapeutic radionuclides such as ^{131}I, ^{90}Y and ^{177}Lu are reported. The red-marrow doses calculated by the traditional and new algorithms are compared for ^{131}I in ablation therapy (14 pts), ^{177}Lu- (13 pts) and ^{90}Y- (11 pts) peptide therapy for neuroendocrine tumours, and ^{90}Y-Zevalin therapy for NHL (21 pts). The range of differences observed is as follows: −36% to −10% for ^{131}I ablation, −22% to 5% for ^{177}Lu-DOTATATE, −9% to 11% for ^{90}Y-DOTATOC and −8% to 6% for ^{90}Y-Zevalin. All differences are mostly due to the activity in the remainder of the body contributing to cross-irradiation. This paper quantifies the influence of mass scaling adjustment on usually applied therapies and shows how to derive the appropriate parameters for other radionuclides and radiopharmaceuticals.

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Introduction

Some models for optimizing radionuclide therapy are based on the maximum tolerable activity to the critical organ (Benua and Leeper 1986, Hänscheid *et al* 2006). The red bone marrow is the critical organ for most radionuclide therapies. Therefore, patient-specific dosimetry needs to be performed in order to determine the maximum administered activity that will not induce significant myelotoxicity (Thomas 2000).

Following the MIRD scheme, the red-marrow absorbed dose in the absence of red-marrow or bone involvement is evaluated by the following equation:

$$D_{\mathrm{rm}} = \tilde{A}_{\mathrm{rm}} \times S_{\mathrm{rm}\leftarrow\mathrm{rm}} + \tilde{A}_{\mathrm{rb}} \times S_{\mathrm{rm}\leftarrow\mathrm{rb}}, \tag{1}$$

where rm (in lower-case letters) is the patient's red marrow, rb (in lower-case letters) is the remainder of the body of the patient, $\tilde{A}$ is the cumulated activity and $S_{\mathrm{target}\leftarrow\mathrm{source}}$ is the MIRD S value for a source irradiating a target (Loevinger *et al* 1988, Stabin *et al* 2002). Note that in this paper the patient's specific parameters are expressed in lower-case letters (e.g. m_{tb} is the total body mass of the patient), while the standard man/woman parameters are expressed in capital letters (e.g. m_{TB} is the total body mass of the standard man/woman). In particular, the notations TB/tb, RB/rb, RM/rm are used to denote reference/patient-specific total body, remainder of body and red-marrow parameters, respectively.

Patient-specific S-factors are usually calculated from the corresponding reference-phantom S-factors (Stabin *et al* 2002, 2001, Wessels *et al* 2004).

The principal equations for patient-specific red-marrow dosimetry are as follows:

$$\tilde{A}_{\mathrm{rm}} = [\tilde{A}_{\mathrm{bl}}] \times m_{\mathrm{rm}} \times \mathrm{RMBLR}, \tag{2}$$

where $[\tilde{A}_{\mathrm{bl}}]$ is the cumulated activity concentration in blood, m_{rm} is the patient's red-marrow mass and RMBLR is the red-marrow-to-blood activity concentration ratio. As regards RMBLR, specific values have been suggested for different therapies. In the case of radioimmunotherapy with monoclonal antibodies, the well-known equation by Sgouros (1993) is generally applied: RMBLR = RMECFF/(1 − HCT), where RMECFF is the patient-specific red-marrow extracellular fluid fraction and HCT is the haematocrit. RMECFF varies among patients. In the absence of specific knowledge a value of 0.19 is generally assumed; thus typical values of RMBLR ranging from 0.2 to 0.4 are derived. In the case of therapies using smaller molecules, such as receptor-binding-peptide radionuclide therapy or ^{131}I therapy for thyroid diseases, the activity concentration in red marrow can be considered equal to the specific activity in blood (i.e. RMBLR = 1) (Sgouros 2005, Cremonesi *et al* 2006).

The biodistribution of radionuclides in the total body includes the activity in the red-marrow, in identified source organs (liver, lungs) and in the remainder of the body. The general equation including all the contributors is reported in appendix A. However, in most cases the contributions to the red-marrow absorbed dose from organs other than red-marrow and remainder of the body are negligibly small, and the following approximations are therefore reasonable:

$$\tilde{A}_{\mathrm{rb}} = \tilde{A}_{\mathrm{tb}} - \tilde{A}_{\mathrm{rm}} = \tilde{A}_{\mathrm{tb}} - [\tilde{A}_{\mathrm{bl}}] \times m_{\mathrm{rm}} \times \mathrm{RMBLR}, \tag{3}$$

$$m_{\mathrm{rb}} = m_{\mathrm{tb}} - m_{\mathrm{rm}}, \tag{4}$$

where m_{rb} is the mass of the remainder of the patient's body.

According to the MIRD formalism (Stabin *et al* 2002, Wessels *et al* 2004, Siegel *et al* 2005),

$$S_{\mathrm{rm}\leftarrow\mathrm{rm}} = S_{\mathrm{RM}\leftarrow\mathrm{RM}} \times \frac{m_{\mathrm{RM}}}{m_{\mathrm{rm}}}, \tag{5}$$

$$S_{\mathrm{rm}\leftarrow\mathrm{rb}} = \left(S_{\mathrm{RM}\leftarrow\mathrm{TB}} \times \frac{m_{\mathrm{tb}}}{m_{\mathrm{rb}}} - S_{\mathrm{RM}\leftarrow\mathrm{RM}} \times \frac{m_{\mathrm{rm}}}{m_{\mathrm{rb}}}\right)\frac{m_{\mathrm{RM}}}{m_{\mathrm{rm}}}. \tag{6}$$

The direct measurement of m_{rm} is not feasible, and thus it is assumed to be scaled linearly with the patient's total body mass m_{tb} (Woodard 1984). This means that m_{rm} can be evaluated as

$$m_{\mathrm{rm}} = m_{\mathrm{RM}}\frac{m_{\mathrm{tb}}}{m_{\mathrm{TB}}}. \tag{7}$$

Substituting equation (2) to (7), equation (1) can be rewritten as

$$D_{\mathrm{rm}} = [\tilde{A}_{\mathrm{bl}}] \times m_{\mathrm{RM}} \times \mathrm{RMBLR} \times S_{\mathrm{RM}\leftarrow\mathrm{RM}} + \left(\tilde{A}_{\mathrm{tb}} - [\tilde{A}_{\mathrm{bl}}] \times m_{\mathrm{RM}}\frac{m_{\mathrm{tb}}}{m_{\mathrm{TB}}} \times \mathrm{RMBLR}\right)$$
$$\times \left(S_{\mathrm{RM}\leftarrow\mathrm{TB}} \times \frac{m_{\mathrm{TB}}}{m_{\mathrm{RB}}} - S_{\mathrm{RM}\leftarrow\mathrm{RM}} \times \frac{m_{\mathrm{RM}}}{m_{\mathrm{RB}}}\right)\frac{m_{\mathrm{TB}}}{m_{\mathrm{tb}}}. \tag{8}$$

Equation (8) is the equation generally used for the patient-specific blood-based red-marrow dose calculation (Stabin *et al* 2002, Wessels *et al* 2004, Siegel *et al* 2005). The first term describes the self-irradiation of the red marrow and the second the cross-irradiation of the red marrow due to the activity in the remainder of the body. Note that in equations (5), (6) and (8), the $S_{\mathrm{TARGET}\leftarrow\mathrm{SOURCE}}$ values are scaled linearly with the ratio of the reference male/female to the patient's organ masses. This gives an approximated value of $S_{\mathrm{target}\leftarrow\mathrm{source}}$ taking into account the real mass of the patient organs which is often very different from the reference male/female. However, more accurate $S_{\mathrm{target}\leftarrow\mathrm{source}}$ factors are not necessarily scaled linearly with mass (Stabin *et al* 2005).

In this paper, the mass-dependent $S_{\mathrm{target}\leftarrow\mathrm{source}}$ values given in the OLINDA/EXM code (Stabin *et al* 2005) were used for the validation of an algorithm for estimating the patient-specific $S_{\mathrm{rm}\leftarrow\mathrm{tb}}$ and $S_{\mathrm{rm}\leftarrow\mathrm{rm}}$ dose conversion factors from the reference male/female $S_{\mathrm{RM}\leftarrow\mathrm{TB}}$ and $S_{\mathrm{RM}\leftarrow\mathrm{RM}}$.

Three isotope-specific parameters, x_1, x_2 and x_3 (see equations (9)–(11)), were evaluated for each of the three widely used therapeutic radionuclides, ^{131}I, ^{90}Y and ^{177}Lu. $S_{\mathrm{rm}\leftarrow\mathrm{tb}}$ and $S_{\mathrm{rm}\leftarrow\mathrm{rm}}$ have been then included in the equation for the red-marrow absorbed dose calculation, obtaining a new equation different from equation (8).

The difference between the red-marrow dose calculated by equation (8) and this new equation was evaluated in patients undergoing different therapeutic approaches using ^{131}I (post-surgical remnant ablation), ^{177}Lu-DOTATATE (neuroendocrine tumours), ^{90}Y-DOTATOC (neuroendocrine tumours) or ^{90}Y-Zevalin (NHL).

Materials and methods

The software OLINDA/EXM offers for each radionuclide the mass-scaled S factors once the patient-specific organ masses are inserted. This allows us to overcome the approximation of $S_{\mathrm{target}\leftarrow\mathrm{source}}$ with $S_{\mathrm{TARGET}\leftarrow\mathrm{SOURCE}}$.

Table 1 presents the $S_{\mathrm{RM}\leftarrow\mathrm{tb}}$ and $S_{\mathrm{RM}\leftarrow\mathrm{RM}}$ values for ^{131}I, ^{90}Y, ^{177}Lu from OLINDA/EXM when different total body masses and standard red marrow (1.12 kg, male; 1.3 kg, female) are considered. Table 2 presents the $S_{\mathrm{rm}\leftarrow\mathrm{TB}}$ and $S_{\mathrm{rm}\leftarrow\mathrm{rm}}$ values for ^{131}I, ^{90}Y, ^{177}Lu from OLINDA/EXM when different red-marrow masses and standard total body weights (73.7 kg, male; 56.912 kg, female) are considered.

The following algorithm is proposed for the $S_{\mathrm{RM}\leftarrow\mathrm{tb}}(m_{\mathrm{tb}}, m_{\mathrm{RM}})$ calculation:

$$S_{\mathrm{RM}\leftarrow\mathrm{tb}}(m_{\mathrm{tb}}, m_{\mathrm{RM}}) = S_{\mathrm{RM}\leftarrow\mathrm{TB}} \times \left(\frac{m_{\mathrm{TB}}}{m_{\mathrm{tb}}}\right)^{x_1}, \tag{9}$$

Table 1. ^{131}I, ^{90}Y and ^{177}Lu $S_{RM\leftarrow tb}$ and $S_{RM\leftarrow RM}$ (m_{tb}) calculated by the OLINDA/EXM code for different total body masses of males and females. The $S_{RM\leftarrow TB}$ and $S_{RM\leftarrow RM}$ values for the reference male and female phantoms are written in bold.

Man			Woman		
m_{tb} (kg)	$S_{(RM\leftarrow tb)}$ (mGy MBq s^{-1})	$S_{(RM\leftarrow RM)}$ (mGy MBq s^{-1})	m_{tb} (kg)	$S_{(RM\leftarrow tb)}$ (mGy MBq s^{-1})	$S_{(RM\leftarrow RM)}$ (mGy MBq s^{-1})
			^{131}I		
50	7.72×10^{-7}	1.55×10^{-5}	40	9.30×10^{-7}	1.41×10^{-5}
60	6.98×10^{-7}	1.55×10^{-5}	50	8.24×10^{-7}	1.41×10^{-5}
70	6.45×10^{-7}	1.55×10^{-5}	**56.912**	$\mathbf{7.72 \times 10^{-7}}$	$\mathbf{1.41 \times 10^{-5}}$
73.7	$\mathbf{6.29 \times 10^{-7}}$	$\mathbf{1.55 \times 10^{-5}}$	60	7.53×10^{-7}	1.41×10^{-5}
80	6.06×10^{-7}	1.55×10^{-5}	70	7.02×10^{-7}	1.41×10^{-5}
90	5.75×10^{-7}	1.55×10^{-5}	80	6.64×10^{-7}	1.41×10^{-5}
100	5.51×10^{-7}	1.55×10^{-5}	90	6.35×10^{-7}	1.41×10^{-5}
			100	6.11×10^{-7}	1.41×10^{-5}
			^{90}Y		
50	2.08×10^{-6}	5.87×10^{-5}	40	2.45×10^{-6}	5.23×10^{-6}
60	1.74×10^{-6}	5.87×10^{-5}	50	1.96×10^{-6}	5.23×10^{-6}
70	1.49×10^{-6}	5.87×10^{-5}	**56.912**	$\mathbf{1.72 \times 10^{-6}}$	$\mathbf{5.23 \times 10^{-6}}$
73.7	$\mathbf{1.41 \times 10^{-6}}$	$\mathbf{5.87 \times 10^{-5}}$	60	1.63×10^{-6}	5.23×10^{-6}
80	1.30×10^{-6}	5.87×10^{-5}	70	1.40×10^{-6}	5.23×10^{-6}
90	1.16×10^{-6}	5.87×10^{-5}	80	1.23×10^{-6}	5.23×10^{-6}
100	1.04×10^{-6}	5.87×10^{-5}	90	1.09×10^{-6}	5.23×10^{-6}
			100	9.80×10^{-7}	5.23×10^{-6}
			^{177}Lu		
50	3.79×10^{-7}	1.19×10^{-5}	40	4.55×10^{-7}	1.05×10^{-5}
60	3.21×10^{-7}	1.19×10^{-5}	50	3.71×10^{-7}	1.05×10^{-5}
70	2.80×10^{-7}	1.19×10^{-5}	**56.912**	$\mathbf{3.30 \times 10^{-7}}$	$\mathbf{1.05 \times 10^{-5}}$
73.7	$\mathbf{2.67 \times 10^{-7}}$	$\mathbf{1.19 \times 10^{-5}}$	60	3.15×10^{-7}	1.05×10^{-5}
80	2.48×10^{-7}	1.19×10^{-5}	70	2.75×10^{-7}	1.05×10^{-5}
90	2.24×10^{-7}	1.19×10^{-5}	80	2.45×10^{-7}	1.05×10^{-5}
100	2.05×10^{-7}	1.19×10^{-5}	90	2.22×10^{-7}	1.05×10^{-5}
			100	2.03×10^{-7}	1.05×10^{-5}

where x_1 is a value which can be obtained by fitting the $S_{RM\leftarrow tb}$ values reported in table 1 with equation (9).

Similarly, $S_{rm\leftarrow TB}(m_{rm}, m_{TB})$ can be calculated as

$$S_{rm\leftarrow TB}(m_{rm}, m_{TB}) = S_{rm\leftarrow TB} \times \left(\frac{m_{RM}}{m_{rm}}\right)^{x_2}, \tag{10}$$

where x_2 is a value which can be obtained by fitting $S_{rm\leftarrow TB}$ reported in table 2 with equation (10).

Finally, the algorithm to calculate $S_{rm\leftarrow rm}(m_{rm})$ is

$$S_{rm\leftarrow rm}(m_{rm}) = S_{RM\leftarrow RM} \times \left(\frac{m_{RM}}{m_{rm}}\right)^{x_3}, \tag{11}$$

where x_3 is a value which can be obtained by fitting the $S_{rm\leftarrow rm}$ values reported in table 2 with equation (11).

Combining equations (9) and (10), the calculation of $S_{rm\leftarrow tb}$ is possible taking into account the true mass of the patient and of the patient's red marrow by

Table 2. ^{131}I, ^{90}Y and ^{177}Lu $S_{rm\leftarrow TB}$ and $S_{rm\leftarrow rm}$ calculated by the OLINDA/EXM code for different red-marrow masses of males and females. The $S_{RM\leftarrow TB}$ and $S_{RM\leftarrow RM}$ values for the reference male and female phantoms are written in bold.

Man			Woman		
m_{rm} (kg)	$S_{(rm\leftarrow TB)}$ (mGy MBq s^{-1})	$S_{(rm\leftarrow rm)}$ (mGy MBq s^{-1})	m_{rm} (kg)	$S_{(rm\leftarrow TB)}$ (mGy MBq s^{-1})	$S_{(rm\leftarrow rm)}$ (mGy MBq s^{-1})
			^{131}I		
0.75	8.78×10^{-7}	2.29×10^{-5}	0.85	1.10×10^{-6}	2.13×10^{-5}
0.85	7.91×10^{-7}	2.03×10^{-5}	1	9.60×10^{-7}	1.82×10^{-5}
0.95	7.21×10^{-7}	1.82×10^{-5}	1.05	9.22×10^{-7}	1.74×10^{-5}
1.05	6.64×10^{-7}	1.65×10^{-5}	1.12	8.74×10^{-7}	1.63×10^{-5}
1.12	$\mathbf{6.29 \times 10^{-7}}$	$\mathbf{1.55 \times 10^{-5}}$	1.2	8.25×10^{-7}	1.53×10^{-5}
1.2	5.95×10^{-7}	1.45×10^{-5}	**1.3**	$\mathbf{7.72 \times 10^{-7}}$	$\mathbf{1.41 \times 10^{-5}}$
1.3	5.57×10^{-7}	1.34×10^{-5}	1.4	7.26×10^{-7}	1.32×10^{-5}
1.5	4.95×10^{-7}	1.17×10^{-5}	1.5	6.86×10^{-7}	1.23×10^{-5}
			^{90}Y		
0.75	2.11×10^{-6}	8.76×10^{-5}	0.85	2.63×10^{-6}	8.00×10^{-5}
0.85	1.86×10^{-6}	7.73×10^{-5}	1	2.24×10^{-6}	6.80×10^{-5}
0.95	1.67×10^{-6}	6.92×10^{-5}	1.05	2.13×10^{-6}	6.47×10^{-5}
1.05	1.51×10^{-6}	6.26×10^{-5}	1.12	2.00×10^{-6}	6.07×10^{-5}
1.12	$\mathbf{1.41 \times 10^{-6}}$	$\mathbf{5.87 \times 10^{-5}}$	1.2	1.87×10^{-6}	5.66×10^{-5}
1.2	1.32×10^{-6}	5.48×10^{-5}	**1.3**	$\mathbf{1.72 \times 10^{-6}}$	$\mathbf{5.23 \times 10^{-5}}$
1.25	1.27×10^{-6}	5.26×10^{-5}	1.4	1.60×10^{-6}	4.85×10^{-5}
1.3	1.22×10^{-6}	5.06×10^{-5}	1.5	1.49×10^{-6}	4.53×10^{-5}
1.4	1.13×10^{-6}	4.69×10^{-5}			
			^{177}Lu		
0.75	3.93×10^{-7}	1.78×10^{-5}	0.85	4.98×10^{-7}	1.60×10^{-5}
0.85	3.48×10^{-7}	1.57×10^{-5}	1	4.26×10^{-7}	1.36×10^{-5}
0.95	3.13×10^{-7}	1.40×10^{-5}	1.05	4.06×10^{-7}	1.30×10^{-5}
1.05	2.84×10^{-7}	1.27×10^{-5}	1.12	3.82×10^{-7}	1.22×10^{-5}
1.12	$\mathbf{2.67 \times 10^{-7}}$	$\mathbf{1.19 \times 10^{-5}}$	1.2	3.57×10^{-7}	1.13×10^{-5}
1.2	2.50×10^{-7}	1.11×10^{-5}	**1.3**	$\mathbf{3.30 \times 10^{-7}}$	$\mathbf{1.05 \times 10^{-5}}$
1.25	2.40×10^{-7}	1.07×10^{-5}	1.4	3.08×10^{-7}	9.73×10^{-6}
1.3	2.31×10^{-7}	1.03×10^{-5}	1.5	2.88×10^{-7}	9.09×10^{-6}
1.4	2.15×10^{-7}	9.53×10^{-5}			
1.5	2.02×10^{-7}	8.90×10^{-5}			

$$S_{rm\leftarrow tb} = S_{RM\leftarrow TB} \times \left(\frac{m_{TB}}{m_{tb}}\right)^{x_1} \left(\frac{m_{RM}}{m_{rm}}\right)^{x_2}, \tag{12}$$

where $(m_{TB}/m_{tb})^{x_1}$ $(m_{RM}/m_{rm})^{x_2}$ represents the final mass scaling factor.

Considering equations (4), (6), (9)–(12), $S_{rm\leftarrow rb}$ can be written as

$$S_{rm\leftarrow rb} = S_{RM\leftarrow TB} \left(\frac{m_{TB}}{m_{rb}}\right)^{x_1} \left(\frac{m_{RM}}{m_{rm}}\right)^{x_2} - S_{RM\leftarrow RM} \times \left(\frac{m_{RM}}{m_{rb}}\right)^{x_3} \left(\frac{m_{RM}}{m_{rm}}\right)^{x_3}. \tag{13}$$

Finally, by equations (1)–(3), (11), (13), equation (1) can be rewritten as

$$D_{rm} = [\tilde{A}_{bl}] \times m_{rm} \times \mathrm{RMBLR} \times S_{RM\leftarrow RM} \times \left(\frac{m_{RM}}{m_{rm}}\right)^{x_3} + (\tilde{A}_{tb} - [\tilde{A}_{bl}] \times m_{rm} \times \mathrm{RMBLR}) \times \left[S_{RM\leftarrow TB} \times \left(\frac{m_{TB}}{m_{rb}}\right)^{x_1} \left(\frac{m_{RM}}{m_{rm}}\right)^{x_2} - S_{RM\leftarrow RM} \times \left(\frac{m_{RM}^2}{m_{rb} m_{rm}}\right)^{x_3}\right]. \tag{14}$$

This represents a new equation for the absorbed red-marrow dose calculation taking into account the patient's specific total body and red-marrow masses, with the self- and cross-irradiation terms. Equation (14) can be used for any known values of m_{rb} and m_{rm}, and is thus completely general.

Inserting equation (7), equation (14) can be rewritten as

$$D_{\mathrm{rm}} = [\tilde{A}_{\mathrm{bl}}] \times m_{\mathrm{RM}} \frac{m_{\mathrm{tb}}}{m_{\mathrm{TB}}} \times \mathrm{RMBLR} \times S_{\mathrm{RM}\leftarrow\mathrm{RM}} \times \left(\frac{m_{\mathrm{TB}}}{m_{\mathrm{tb}}}\right)^{x_3} + \left(\tilde{A}_{\mathrm{tb}} - [\tilde{A}_{\mathrm{bl}}] \times m_{\mathrm{RM}} \frac{m_{\mathrm{tb}}}{m_{\mathrm{TB}}} \times \mathrm{RMBLR}\right) \times \left[S_{\mathrm{RM}\leftarrow\mathrm{TB}} \times \left(\frac{m_{\mathrm{TB}}^2}{m_{\mathrm{tb}} m_{\mathrm{RB}}}\right)^{x_1} \left(\frac{m_{\mathrm{TB}}}{m_{\mathrm{tb}}}\right)^{x_2} - S_{\mathrm{RM}\leftarrow\mathrm{RM}} \times \left(\frac{m_{\mathrm{RM}} m_{\mathrm{TB}}^2}{m_{\mathrm{tb}}^2 m_{\mathrm{RB}}}\right)^{x_3}\right]. \quad (15)$$

The patient-specific red-marrow absorbed dose, D_{rm}, thus depends on the mass of the patient m_{tb}, on x_1, x_2 and x_3, specific for the radionuclide, and on the standard $S_{\mathrm{RM}\leftarrow\mathrm{TB}}$, $S_{\mathrm{RM}\leftarrow\mathrm{RM}}$ values available in the literature (RAdiation Dose Assessment Resource (RADAR)).

To quantify in clinical practice, the impact of the corrections described x_1, x_2 and x_3 were evaluated for ^{131}I, ^{90}Y and ^{177}Lu (table 3), respectively, which are widely used for therapeutic purposes. The red-marrow dose, according to equations (8) and (15), was evaluated in 59 patients undergoing different therapeutic procedures: ^{131}I for remnant ablation therapy (14 pts; 3 m, 11 f); ^{177}Lu- (13 pts; 6 m, 7 f) and ^{90}Y-peptide therapy for neuroendocrine tumours (11 pts; 6 m, 5 f) and ^{90}Y-Zevalin therapy for NHL (21 pts; 16 m, 5 f). A comparison between linear (equation (8)) and nonlinear (equation (15)) mass scaling factors was evaluated in terms of red-marrow dose differences.

In the case of ^{131}I, the activity in the remainder of the body was obtained by subtracting the activity in the red marrow from that in the total body (equation (3), table 4). For ^{177}Lu and ^{90}Y, the activity in the remainder of the body was obtained by subtracting the sum of the activities in the identified organs (e.g. liver, kidneys, spleen) from the total body activity (tables 5–7).

For ^{131}I and ^{177}Lu-/^{90}Y-peptide therapies the RMBLR factor was considered to be 1, while for ^{90}Y-Zevalin, using radiolabelled monoclonal antibodies, the patient-specific RMBLR value was evaluated based on haematocrit and body mass (Sgouros 2005).

Results

In table 3, x_1, x_2, x_3 values are reported for ^{131}I, ^{90}Y and ^{177}Lu, respectively, for male and female subjects. The accuracy of $S_{\mathrm{RM}\leftarrow\mathrm{tb}}$ (equation (9)), $S_{\mathrm{rm}\leftarrow\mathrm{TB}}$ (equation (10)) and $S_{\mathrm{rm}\leftarrow\mathrm{rm}}$ (equation (11)) calculated by linear ($x_1 = x_2 = x_3 = 1$) and nonlinear (x_1, x_2, x_3 from table 3) scaling of $S_{\mathrm{RM}\leftarrow\mathrm{TB}}$ is shown in figures 1–3. The S-factors by OLINDA are considered to be the gold standard.

In table 4, the results comparing the traditional (equation (8)) and the new (equation (15)) equations for the red-marrow doses per unit administered activity (mGy MBq^{-1}) are presented for patients treated with ^{131}I.

Differences (%) range between −36% and −10% mostly due to differences in the cross-irradiation term (−81%, −18%). In contrast, differences in self-irradiation are negligible, with a range of −1%, 2%. On average, the contribution of the cross-irradiation to the total red-marrow dose is 45%.

Tables 5–7 report the same comparison for patients treated with ^{177}Lu-DOTATATE, ^{90}Y-DOTATOC and ^{90}Y-Zevalin, respectively. Differences (%) for ^{177}Lu-DOTATATE range

Table 3. Scaling factors for male and female, for ^{131}I, ^{90}Y and ^{177}Lu. $R^2 = 1.000$ means that $R^2 > 0.9995$. The same parameters for adolescents/children are reported in appendix B.

	Man						Woman					
	^{131}I	R^2	^{90}Y	R^2	^{177}Lu	R^2	^{131}I	R^2	^{90}Y	R^2	^{177}Lu	R^2
x_1	0.50 ± 0.02	0.993	1.003 ± 0.005	1.000	0.896 ± 0.005	1.000	0.47 ± 0.02	0.990	0.999 ± 0.003	1.000	0.894 ± 0.008	1.000
x_2	0.831 ± 0.002	1.000	1.005 ± 0.005	1.000	0.9632 ± 0.0009	1.000	0.8323 ± 0.0006	1.000	1.002 ± 0.002	1.000	0.970 ± 0.002	1.000
x_3	0.974 ± 0.001	1.000	0.9986 ± 0.0006	1.000	1.001 ± 0.003	1.000	0.972 ± 0.004	1.000	1.0000 ± 0.0009	1.000	0.992 ± 0.004	1.000

Table 4. A comparison between red-marrow doses per administered activity (mGy MBq^{-1}) calculated by the classical equation (8) and the new equation (15) in 14 patients submitted to the ^{131}I ablative therapy. Note that the difference ranges between −36% and −10%. Almost all the difference depends on the cross-irradiation due to the activity present in the remainder of the body.

					Total			Self-irradiation			Cross-irradiation		
Patient no.	Sex	$\tilde{A}_{TB}$ (h)	$[\tilde{A}_{bl}]$ (h/kg)	m_p (kg)	D_{RM} (mGy MBq^{-1}) by equation (15)	D_{RM} (mGy MBq^{-1}) by equation (8)	Diff.%	D_{RM} (mGy MBq^{-1}) by equation (15)	D_{RM} (mGy MBq^{-1}) by equation (8)	Diff.%	D_{RM} (mGy MBq^{-1}) by equation (15)	D_{RM} (mGy MBq^{-1}) by equation (8)	Diff.%
1	F	37	0.55	53	0.079	0.101	−28	0.036	0.036	0	0.043	0.065	−52
2	M	29	0.38	85	0.052	0.060	−15	0.024	0.024	0	0.028	0.036	−28
3	M	25	0.70	70	0.071	0.080	−14	0.044	0.044	0	0.027	0.037	−36
4	M	26	0.65	73	0.067	0.076	−14	0.040	0.040	0	0.027	0.036	−34
5	F	31	0.70	58	0.081	0.095	−18	0.046	0.046	0	0.034	0.049	−43
6	F	21	0.28	90	0.036	0.040	−10	0.019	0.019	1	0.018	0.021	−21
7	F	22	0.64	57	0.066	0.077	−16	0.042	0.042	0	0.024	0.035	−45
8	F	14	0.37	51	0.041	0.051	−23	0.024	0.024	0	0.017	0.026	−57
9	F	20	0.41	70	0.047	0.053	−12	0.027	0.027	1	0.020	0.026	−31
10	F	67	0.45	105	0.080	0.089	−11	0.030	0.030	2	0.050	0.059	−18
11	F	15	0.62	60	0.057	0.063	−11	0.041	0.041	0	0.016	0.022	−40
12	F	36	0.41	78	0.061	0.070	−14	0.027	0.027	1	0.034	0.042	−26
13	F	22	0.53	75	0.056	0.061	−10	0.035	0.035	1	0.021	0.026	−27
14	F	33	0.75	44	0.087	0.119	−36	0.049	0.049	−1	0.039	0.070	−81

Table 5. A comparison between red-marrow doses per administered activity (mGy MBq^{-1}) calculated by the classical equation (8) and the new equation (15) in 13 patients treated with ^{177}Lu-DOTATATE therapy for neuroendocrine tumours. Note that the difference ranges between −22% and 5%. All the difference depends on the cross-irradiation due to the activity present in the remainder of the body.

						Total			Self-irradiation			Cross-irradiation		
Patient no.	Sex	$\tilde{A}_{TB}$ (h)	$\tilde{A}_{RB}$ (h)	$[\tilde{A}_{bl}]$ (h/kg)	m_p (kg)	D_{RM} (mGy MBq^{-1}) by equation (15)	D_{RM} (mGy MBq^{-1}) by equation (8)	Diff.%	D_{RM} (mGy MBq^{-1}) by equation (15)	D_{RM} (mGy MBq^{-1}) by equation (8)	Diff.%	D_{RM} (mGy MBq^{-1}) by equation (15)	D_{RM} (mGy MBq^{-1}) by equation (8)	Diff.%
1	M	39	20	0.24	73	0.018	0.018	1	0.012	0.012	0	0.006	0.006	3
2	M	32	26	0.23	96	0.017	0.017	−5	0.011	0.011	0	0.005	0.006	−14
3	M	28	20	0.34	72	0.023	0.023	1	0.017	0.017	0	0.007	0.006	3
4	M	37	27	0.35	62	0.028	0.027	5	0.017	0.017	0	0.012	0.010	11
5	M	25	9	0.19	70	0.012	0.012	1	0.009	0.009	0	0.003	0.003	5
6	M	24	15	0.13	80	0.011	0.011	−1	0.006	0.006	0	0.004	0.004	−3
7	F	22	18	0.23	57	0.016	0.017	−11	0.011	0.011	0	0.004	0.006	−39
8	F	57	48	0.14	50	0.02	0.025	−22	0.007	0.007	0	0.014	0.018	−33
9	F	45	31	0.40	59	0.027	0.029	−11	0.020	0.020	0	0.007	0.010	−41
10	F	26	18	0.27	44	0.019	0.021	−9	0.013	0.013	0	0.006	0.008	−27
11	F	35	24	0.39	48	0.027	0.029	−9	0.019	0.019	0	0.007	0.010	−31
12	F	31	21	0.39	48	0.026	0.028	−8	0.019	0.019	0	0.006	0.008	−31
13	F	34	22	0.13	58	0.011	0.013	−18	0.006	0.006	0	0.005	0.007	−40

Table 6. A comparison between red-marrow doses per unit administered activity (mGy MBq^{-1}) calculated by the classical equation (8) and the new equation (15) in 11 patients treated with ^{90}Y-DOTATOC therapy for neuroendocrine tumours. Note that the difference ranges between −9% and 11%. All the difference depends on the cross-irradiation due to the activity present in the remainder of the body.

						Total			Self-irradiation			Cross-irradiation		
Patient no.	Sex	$\tilde{A}_{TB}$ (h)	$\tilde{A}_{RB}$ (h)	$[\tilde{A}_{bl}]$ (h/kg)	m_p (kg)	D_{RM} (mGy MBq^{-1}) by equation (15)	D_{RM} (mGy MBq^{-1}) by equation (8)	Diff.%	D_{RM} (mGy MBq^{-1}) by equation (15)	D_{RM} (mGy MBq^{-1}) by equation (8)	Diff.%	D_{RM} (mGy MBq^{-1}) by equation (15)	D_{RM} (mGy MBq^{-1}) by equation (8)	Diff.%
1	M	12	6.06	0.07	80	0.025	0.026	−5	0.016	0.016	0	0.009	0.011	−12
2	M	11	5.32	0.16	76	0.047	0.047	−1	0.038	0.038	0	0.009	0.010	−7
3	M	32	22.90	0.16	76	0.077	0.080	−3	0.038	0.038	0	0.039	0.042	−7
4	M	17	8.65	0.07	87	0.027	0.029	−9	0.015	0.015	0	0.011	0.014	−22
5	M	22	13.40	0.09	80	0.042	0.045	−6	0.021	0.021	0	0.021	0.023	−12
6	M	18	8.77	0.06	85	0.026	0.028	−9	0.013	0.013	0	0.012	0.014	−20
7	F	21	9.03	0.15	45	0.065	0.059	9	0.037	0.037	0	0.028	0.022	21
8	F	19	7.00	0.12	51	0.047	0.046	4	0.030	0.030	0	0.017	0.015	10
9	F	21	9.09	0.10	45	0.052	0.046	11	0.024	0.024	0	0.028	0.022	21
10	F	29	18.00	0.20	65	0.076	0.080	−5	0.049	0.049	0	0.027	0.031	−14
11	F	21	4.82	0.08	60	0.028	0.028	−2	0.019	0.019	0	0.008	0.009	−5

Table 7. A comparison between red-marrow doses per administered activity (mGy MBq^{-1}) calculated by the classical equation (8) and the new equation (15) in 21 patients treated with ^{90}Y-Zevalin therapy for NHL. Note that the difference ranges between −8% and 6%. All the difference depends on the cross-irradiation due to the activity present in the remainder of the body. Note that in this case, RMBLR has been calculated for each patient.

							Total			Self-irradiation			Cross-irradiation		
Patient no.	Sex	RMECFF/ (1 − HCT)	$\tilde{A}_{TB}$ (h)	$\tilde{A}_{RB}$ (h)	$[\tilde{A}_{bl}]$ (h/kg)	m_p (kg)	D_{RM} (mGy MBq^{-1}) by equation (15)	D_{RM} (mGy MBq^{-1}) by equation (8)	Diff.%	D_{RM} (mGy MBq^{-1}) by equation (15)	D_{RM} (mGy MBq^{-1}) by equation (8)	Diff.%	D_{RM} (mGy MBq^{-1}) by equation (15)	D_{RM} (mGy MBq^{-1}) by equation (8)	Diff.%
1	M	0.30	84	67.20	5.656 09	86	0.496	0.515	−4	0.406	0.406	0	0.09	0.109	−21
2	M	0.30	85	58.30	9.414 47	84	0.748	0.763	−2	0.666	0.666	0	0.082	0.097	−18
3	M	0.31	88	68.80	5.116 29	108	0.437	0.468	−7	0.379	0.379	0	0.058	0.089	−53
4	M	0.30	83	42.70	9.673 91	66	0.796	0.789	1	0.698	0.698	0	0.098	0.09	8
5	M	0.30	87	67.00	9.181 29	99	0.725	0.752	−4	0.658	0.657	0	0.068	0.094	−40
6	M	0.29	68	50.10	8.655 44	74	0.686	0.690	0	0.595	0.595	0	0.091	0.094	−4
7	M	0.32	73	41.70	3.712 50	69	0.368	0.365	1	0.281	0.281	0	0.087	0.084	4
8	M	0.28	89	58.90	8.907 42	69	0.722	0.718	1	0.599	0.599	0	0.123	0.119	4
9	M	0.28	89	71.10	7.702 42	84	0.619	0.637	−3	0.519	0.519	0	0.100	0.118	−18
10	M	0.32	79	60.80	8.812 45	74	0.783	0.786	0	0.671	0.671	0	0.112	0.115	−3
11	M	0.32	79	55.12	8.671 03	86	0.733	0.748	−2	0.659	0.659	0	0.074	0.089	−21
12	M	0.32	83	68.41	5.889 50	95	0.526	0.551	−5	0.451	0.450	0	0.075	0.100	−34
13	M	0.26	65	46.80	8.742 36	63	0.659	0.645	2	0.541	0.542	0	0.118	0.104	12
14	M	0.25	77	52.24	6.346 60	70	0.483	0.481	0	0.377	0.377	0	0.106	0.104	2
15	M	0.26	74	55.18	7.922 05	55	0.672	0.629	6	0.488	0.489	0	0.183	0.14	24
16	M	0.26	84	63.75	10.221 82	80	0.733	0.745	−2	0.633	0.633	0	0.100	0.111	−12
17	F	0.28	82	65.00	8.958 42	63	0.709	0.720	−2	0.606	0.606	0	0.103	0.114	−11
18	F	0.28	70	42.30	6.555 60	66	0.517	0.527	−2	0.456	0.456	0	0.061	0.071	−16
19	F	0.26	68	47.90	4.996 92	67	0.387	0.399	−3	0.321	0.321	0	0.067	0.079	−18
20	F	0.27	69	53.00	4.499 27	100	0.332	0.357	−8	0.299	0.299	0	0.033	0.058	−76
21	F	0.28	86	65.00	7.865 62	58	0.663	0.665	0	0.542	0.542	0	0.121	0.123	−2

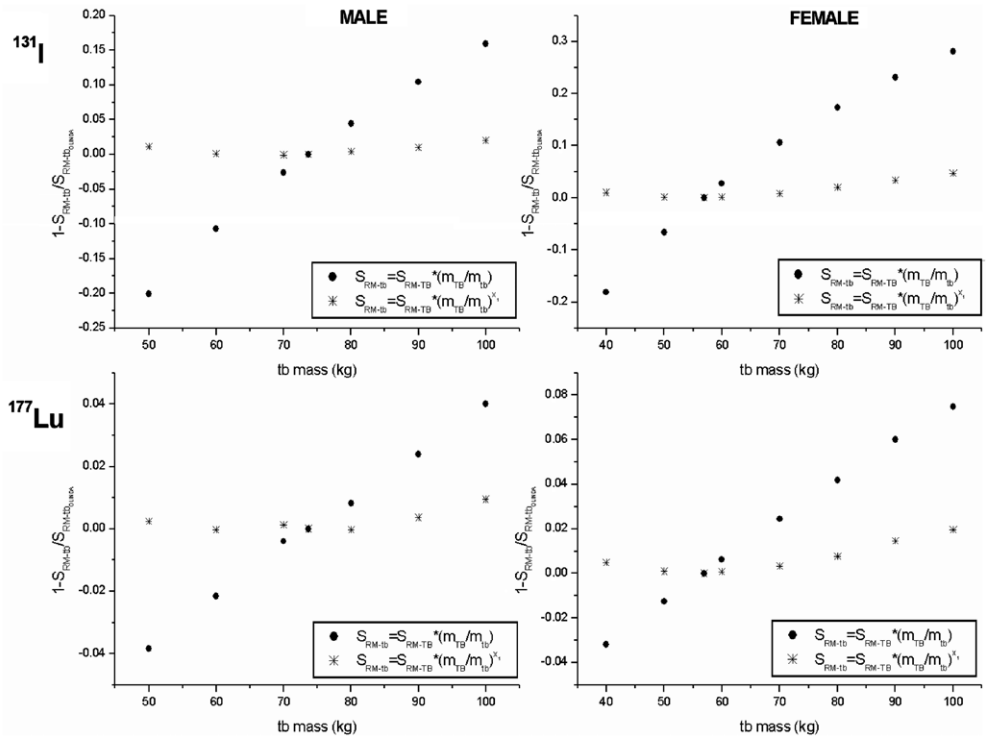

Figure 1. Accuracy of $S_{\mathrm{RM}\leftarrow\mathrm{tb}}$ calculated by linear (equation (9) with $x_1 = 1$) and nonlinear (equation (9) with x_1 from table 3) scaling of $S_{\mathrm{RM}\leftarrow\mathrm{TB}}$. The $S_{\mathrm{RM}\leftarrow\mathrm{tb}}$ factors calculated by OLINDA are considered to be the gold standard. The uncertainties are shown for ^{131}I and ^{177}Lu. Those calculated for ^{90}Y are less than 0.005 by both methods.

between −22% and 5%, with an average cross-irradiation contribution of 34% to the red-marrow dose (table 5), while differences (%) for ^{90}Y-DOTATOC vary between −9% and 11%, with an average cross-irradiation contribution of 41% to the red-marrow dose (table 6). Finally, for ^{90}Y-Zevalin the range of differences (%) is −8% to 6%, and the average contribution of the cross-irradiation is 15% (table 7). In all cases, differences are related to the cross-irradiation term only.

Discussion

In performing the patient-specific blood-based red-marrow dosimetry, a scaling of the dose conversion factors for the reference phantoms (the $S_{\mathrm{TARGET}\leftarrow\mathrm{SOURCE}}$ values in the MIRD/RADAR system) may be needed. This allows one to obtain patient-specific $S_{\mathrm{target}\leftarrow\mathrm{source}}$ values based on the $S_{\mathrm{TARGET}\leftarrow\mathrm{SOURCE}}$ for reference phantoms. In the majority of cases, when no direct involvement of red marrow or bone occurs, there are only two S-factors to take into account: $S_{\mathrm{rm}\leftarrow\mathrm{rm}}$ and $S_{\mathrm{rm}\leftarrow\mathrm{tb}}$.

Generally, a linear scaling factor for $S_{\mathrm{RM}\leftarrow\mathrm{RM}}$ and $S_{\mathrm{RM}\leftarrow\mathrm{TB}}$ has been proposed (see equations (5), (6), (8)) (Stabin *et al* 2002, Wessels *et al* 2004, Siegel *et al* 2005, Shen *et al* 1999).

In a recently published paper, Siegel and Stabin illustrated that each radionuclide may have its own mass scaling factor, that this scaling factor may not be necessarily linear and that

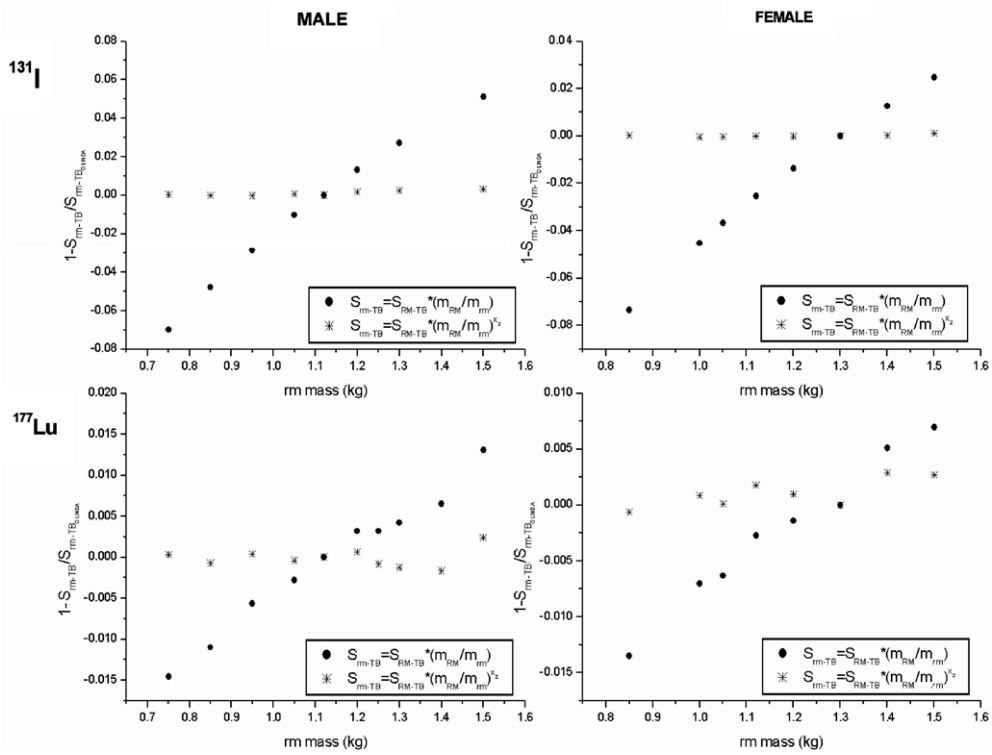

Figure 2. Accuracy of $S_{\mathrm{rm}\leftarrow\mathrm{TB}}$ calculated by linear (equation (10) with $x_2 = 1$) and nonlinear (equation (10) with x_2 from table 3) scaling of $S_{\mathrm{RM}\leftarrow\mathrm{TB}}$. The $S_{\mathrm{rm}\leftarrow\mathrm{TB}}$ factors calculated by OLINDA are considered to be the gold standard. The uncertainties are shown for ^{131}I and ^{177}Lu. Those calculated for ^{90}Y are less than 0.005 by both methods.

the cross-dose term may be corrected taking into account the appropriate mass scaling factor (Siegel and Stabin 2007). The software code OLINDA/EXM has replaced the MIRDOSE code which was previously widely used and cited. It has been approved by the US Food and Drug Administration and it is also used in many centres for internal dosimetry calculation. OLINDA/EXM has a new, useful facility that allows modification of the standard model reference organ masses to represent the actual masses of individual patient organs. The modifications that it makes are approximate, assuming linear scaling of organ self-dose for electrons and scaling to the 1/3 power for photons. A better approach would be to perform new Monte Carlo studies with models of different sizes and see the real results with all changes in geometry and radiation transport that will occur in different sized individuals. Nonetheless, this facility is useful for studying some simple relationships that can affect individual subject dosimetry, if the uncertainties of the method (and of all other variables involved in the problem) are borne in mind.

Once the correct mass scaling factors for the *S* dose coefficients are obtained (by the algorithm presented as well as by OLINDA/EXM), a suitable equation for red-marrow dose calculation must be applied. Starting from the $S_{\mathrm{rm}\leftarrow\mathrm{tb}}$ and $S_{\mathrm{rm}\leftarrow\mathrm{rm}}$ values for different total body and red-marrow masses reported by OLINDA/EXM, a new equation for the patient-specific red-marrow blood-based dose calculation has been developed including x_1, x_2 and x_3. The values x_1, x_2 and x_3 were calculated specifically for three radionuclides (^{131}I, ^{177}Lu and ^{90}Y)

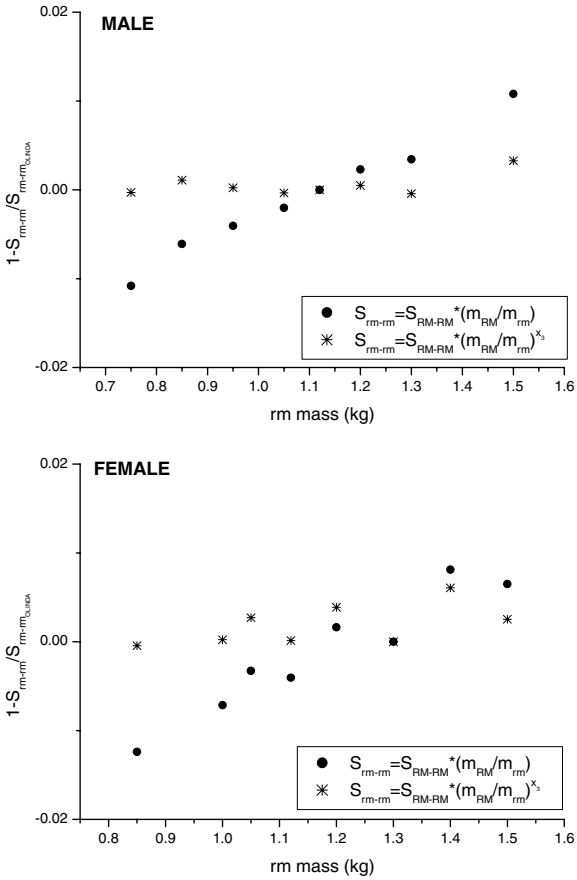

Figure 3. Accuracy of $S_{\mathrm{rm}\leftarrow\mathrm{rm}}$ calculated by linear (equation (11) with $x_3 = 1$) and nonlinear (equation (11) with x_3 from table 3) scaling of $S_{\mathrm{RM}\leftarrow\mathrm{RM}}$. The $S_{\mathrm{rm}\leftarrow\mathrm{rm}}$ factors calculated by OLINDA are considered to be the gold standard. The uncertainties are shown only for ^{131}I. Those calculated for ^{90}Y and ^{177}Lu are less than 0.005 by both methods.

usually employed in radionuclide therapy, and the red-marrow doses were evaluated by the traditional (equation (8)) and the new (equation (15)) algorithms. Non-negligible differences emerge between the results obtained by the two methods (tables 4–7), mostly due to the activity in the remainder of the body (cross-irradiation).

Certainly, the quantitative impact of using the appropriate *S*-factor values for calculation of the red-marrow dose depends on the physical characteristics of the radionuclide and the pharmacokinetics of the radiopharmaceutical. For these reason, different therapies were considered to evaluate the influence of mass scaling adjustment.

The gamma emission of ^{131}I as well as the characteristic pharmacokinetics of iodide serves to increase the contribution of the cross-dose term to the red-marrow dose and yield a relatively large difference between the traditional and new red-marrow dose estimates. ^{177}Lu is a gamma emitter but of much lower gamma abundance and energy compared to ^{131}I, leading to a minor cross-dose contribution. Moreover, the fast clearance from the blood typical of peptides lowers the impact on the total red-marrow dose evaluated by the new algorithm (equation (15)). For ^{90}Y-conjugates the cross-irradiation contribution is only due to *bremsstrahlung*, and the impact

of the cross-irradiation on the total dose is very much dependent on the blood clearance. In the case of ^{90}Y-peptides, the fast clearance allows a low total dose with similar contributions from self- and cross-terms, while in the case of ^{90}Y-Zevalin, characterized by a slow blood clearance, high values of total red-marrow dose are usually obtained with a minor contribution of cross-irradiation. Consequently, differences in the cross-term are approximately halved in the case of ^{90}Y-peptides, while even large differences in the cross-dose evaluation (see patient 20, table 7) are smoothed once incorporated in the total dose for ^{90}Y-Zevalin.

Conclusion

This paper demonstrates that a radionuclide-dependent, nonlinear $S_{\mathrm{TARGET}\leftarrow\mathrm{SOURCE}}$ scaling factor must replace the linear approximation to obtain a more accurate value of red-marrow absorbed dose. Moreover, this nonlinear scaling factor must be applied not only to the self-irradiation term (as usually proposed by the traditional algorithm and applied by OLINDA/EXM), but also to the cross-irradiation term, which is largely responsible for the difference between red-marrow doses calculated by the traditional and the new algorithms. If the mass-dependent dose coefficients (*S*-factors) are those reported by OLINDA/EXM, the scaling factors can be obtained by equations (9)–(12), and consequently the new general equation for the red-marrow dose calculation must be used (equation (14)). Although the differences in the red-marrow doses calculated by the old and new algorithms are admittedly small compared to other possible uncertainties in the dosimetric analysis (activity quantification, kinetic parameter measurements, organ mass values), we agree with the opinion of Siegel and Stabin that the most accurate values should always be used to minimize the overall error in the calculation (Siegel and Stabin 2007).

Appendix A

In some cases, more than two source organs (tb and rm) are involved in the red-marrow dosimetry. In this case, the contribution of organs other than total body and red-marrow itself must be included in the calculation of the red-marrow absorbed dose. Thus, equation (14) has to be replaced by the more general equation:

$$D_{\mathrm{rm}} = \tilde{A}_{\mathrm{rm}} \times S_{\mathrm{rm}\leftarrow\mathrm{rm}} + \sum_{\mathrm{h}} \tilde{A}_{\mathrm{h}} \times S_{\mathrm{rm}\leftarrow\mathrm{h}} + \tilde{A}_{\mathrm{rb}} \times S_{\mathrm{rm}\leftarrow\mathrm{rb}}, \tag{A.1}$$

where 'h' is the organs contributing to the red-marrow absorbed dose.

$$S_{\mathrm{rm}\leftarrow\mathrm{h}} = S_{\mathrm{RM}\leftarrow\mathrm{H}} \times \left(\frac{m_{\mathrm{H}}}{m_{\mathrm{h}}}\right)^{x_{\mathrm{RM}\leftarrow\mathrm{h}}} \left(\frac{m_{\mathrm{RM}}}{m_{\mathrm{rm}}}\right)^{x_{\mathrm{rm}\leftarrow\mathrm{H}}}, \tag{A.2}$$

where 'H' means the 'h' organ of the male/female reference phantom.

$x_{\mathrm{RM}\leftarrow\mathrm{h}}$ must be calculated by fitting $S_{\mathrm{RM}\leftarrow\mathrm{h}}\,(m, m_{\mathrm{RM}})$ with

$$S_{\mathrm{RM}\leftarrow\mathrm{h}}(m, m_{\mathrm{RM}}) = S_{\mathrm{RM}\leftarrow\mathrm{H}} \times \left(\frac{m_{\mathrm{H}}}{m_{\mathrm{h}}}\right)^{x_{\mathrm{RM}\leftarrow\mathrm{h}}}, \tag{A.3}$$

and $x_{\mathrm{rm}\leftarrow\mathrm{H}}$ can be calculated by fitting $S_{\mathrm{rm}\leftarrow\mathrm{H}}(m, m_{\mathrm{H}})$ with

$$S_{\mathrm{rm}\leftarrow\mathrm{H}}(m, m_{\mathrm{H}}) = S_{\mathrm{RM}\leftarrow\mathrm{H}} \times \left(\frac{m_{\mathrm{RM}}}{m_{\mathrm{rm}}}\right)^{x_{\mathrm{rm}\leftarrow\mathrm{H}}}. \tag{A.4}$$

The cumulated activity in the remainder of the body is

$$\tilde{A}_{\mathrm{rb}} = \tilde{A}_{\mathrm{tb}} - \tilde{A}_{\mathrm{rm}} - \sum_{\mathrm{h}} \tilde{A}_{\mathrm{h}}. \tag{A.5}$$

Table B1. m_{TB}, m_{RM}, $S_{RM\leftarrow TB}$ and $S_{RM\leftarrow RM}$ for adolescent/children reference phantoms.

			^{131}I		^{90}Y		^{177}Lu	
Phantom	m_{TB} (kg)	m_{RM} (kg)	$S_{RM\leftarrow TB}$ (mGy MBq s^{-1})	$S_{RM\leftarrow RM}$ (mGy MBq s^{-1})	$S_{RM\leftarrow TB}$ (mGy MBq s^{-1})	$S_{RM\leftarrow RM}$ (mGy MBq s^{-1})	$S_{RM\leftarrow TB}$ (mGy MBq s^{-1})	$S_{RM\leftarrow RM}$ (mGy MBq s^{-1})
15 years	56.8	1.05	7.99×10^{-7}	1.71×10^{-5}	1.90×10^{-6}	6.47×10^{-5}	3.50×10^{-7}	1.29×10^{-5}
10 years	33.2	0.61	1.28×10^{-6}	2.93×10^{-5}	3.35×10^{-6}	1.13×10^{-4}	6.10×10^{-7}	2.28×10^{-5}
5 years	19.8	0.32	2.18×10^{-6}	6.00×10^{-5}	6.53×10^{-6}	2.36×10^{-4}	1.15×10^{-6}	4.81×10^{-5}
1 year	9.72	0.15	5.68×10^{-6}	1.48×10^{-4}	2.18×10^{-6}	5.75×10^{-4}	3.42×10^{-6}	1.21×10^{-4}
Newborn	3.6	0.047	2.30×10^{-6}	5.10×10^{-4}	1.11×10^{-4}	1.99×10^{-3}	1.50×10^{-5}	4.20×10^{-4}

Table B2. Scaling factors for adolescents/children, for ^{131}I, ^{90}Y and ^{177}Lu. $R^2 = 1.000$ means that $R^2 > 0.9995$.

	^{131}I	R^2	^{90}Y	R^2	^{177}Lu	R^2
			15 years old			
x_1	0.53 ± 0.01	0.996	1.002 ± 0.002	1.000	0.910 ± 0.006	1.000
x_2	0.838 ± 0.001	1.000	1.000 ± 0.003	1.000	0.969 ± 0.002	1.000
x_3	0.977 ± 0.005	1.000	1.001 ± 0.001	1.000	1.003 ± 0.007	1.000
			10 years old			
x_1	0.58 ± 0.02	0.995	1.001 ± 0.001	1.000	0.922 ± 0.003	1.000
x_2	0.855 ± 0.004	1.000	1.004 ± 0.002	1.000	0.9720 ± 0.0009	1.000
x_3	0.978 ± 0.003	1.000	1.009 ± 0.005	1.000	1.002 ± 0.002	1.000
			5 years old			
x_1	0.65 ± 0.01	0.998	1.000 ± 0.001	1.000	0.949 ± 0.004	1.000
x_2	0.879 ± 0.004	1.000	0.9980 ± 0.0009	1.000	1.02 ± 0.05	0.985
x_3	0.9886 ± 0.0007	1.000	0.996 ± 0.001	1.000	1.04 ± 0.05	0.985
			1 year old			
x_1	0.783 ± 0.007	1.000	1.001 ± 0.002	1.000	0.964 ± 0.002	1.000
x_2	0.9234 ± 0.0003	1.000	1.000 ± 0.002	1.000	0.985 ± 0.001	1.000
x_3	0.999 ± 0.004	1.000	1.0015 ± 0.0004	1.000	1.008 ± 0.004	1.000
			Newborn			
x_1	0.894 ± 0.003	1.000	0.988 ± 0.007	1.000	0.984 ± 0.004	1.000
x_2	0.961 ± 0.002	1.000	0.989 ± 0.006	1.000	0.995 ± 0.004	1.000
x_3	0.995 ± 0.001	1.000	0.998 ± 0.002	1.000	0.998 ± 0.001	1.000

The dose coefficient $S_{\mathrm{rm}\leftarrow\mathrm{rb}}$ is

$$S_{\mathrm{rm}\leftarrow\mathrm{rb}} = S_{\mathrm{RM}\leftarrow\mathrm{TB}} \times \left(\frac{m_{\mathrm{TB}}}{m_{\mathrm{tb}}}\right)^{x_1} \left(\frac{m_{\mathrm{RM}}}{m_{\mathrm{rm}}}\right)^{x_2} - S_{\mathrm{RM}\leftarrow\mathrm{RM}} \times \left(\frac{m_{\mathrm{RM}}}{m_{\mathrm{rb}}}\right)^{x_3} \left(\frac{m_{\mathrm{RM}}}{m_{\mathrm{rm}}}\right)^{x_3} - \sum_{\mathrm{h}} S_{\mathrm{RM}\leftarrow\mathrm{H}} \times \left(\frac{m_{\mathrm{H}}}{m_{\mathrm{rb}}}\right)^{x_{\mathrm{RM}\leftarrow\mathrm{h}}} \left(\frac{m_{\mathrm{RM}}}{m_{\mathrm{rm}}}\right)^{x_{\mathrm{rm}\leftarrow\mathrm{H}}}. \tag{A.6}$$

Note that

$$m_{\mathrm{rb}} = m_{\mathrm{tb}} - m_{\mathrm{rm}} - \sum_{\mathrm{h}} m_{\mathrm{h}}. \tag{A.7}$$

From equations (A.1), (A.2) and (A.6), D_{rm} must be calculated as

$$D_{\mathrm{rm}} = \tilde{A}_{\mathrm{rm}} S_{\mathrm{RM}\leftarrow\mathrm{RM}} \times \left(\frac{m_{\mathrm{RM}}}{m_{\mathrm{rm}}}\right)^{x_3} + \sum \tilde{A}_{\mathrm{h}} S_{\mathrm{RM}\leftarrow\mathrm{H}} \times \left(\frac{m_{\mathrm{H}}}{m_{\mathrm{h}}}\right)^{x_{\mathrm{RM}\leftarrow\mathrm{h}}} \left(\frac{m_{\mathrm{RM}}}{m_{\mathrm{rm}}}\right)^{x_{\mathrm{rm}\leftarrow\mathrm{H}}} + \tilde{A}_{\mathrm{rb}} \left[S_{\mathrm{RM}\leftarrow\mathrm{TB}} \left(\frac{m_{\mathrm{TB}}}{m_{\mathrm{rb}}}\right)^{x_1} \left(\frac{m_{\mathrm{RM}}}{m_{\mathrm{rm}}}\right)^{x_2} - S_{\mathrm{RM}\leftarrow\mathrm{RM}} \times \left(\frac{m_{\mathrm{RM}}^2}{m_{\mathrm{rb}} m_{\mathrm{rm}}}\right)^{x_3} - \sum S_{\mathrm{RM}\leftarrow\mathrm{H}} \times \left(\frac{m_{\mathrm{H}}}{m_{\mathrm{rb}}}\right)^{x_{\mathrm{RM}\leftarrow\mathrm{h}}} \left(\frac{m_{\mathrm{RM}}}{m_{\mathrm{rm}}}\right)^{x_{\mathrm{rm}\leftarrow\mathrm{H}}} \right]. \tag{A.8}$$

This is the general equation for the red-marrow absorbed dose calculation when more than two organs (compartments) are involved.

Appendix B

m_{TB}, m_{RM}, $S_{RM \leftarrow TB}$ and $S_{RM \leftarrow RM}$ for the 15 years, 10 years, 5 years, 1 year old and newborn reference phantoms are reported in table B1. Although the radionuclide therapies are usually employed only in adult patients, for completeness, x_1, x_2 and x_3 were calculated by equations (9)–(11), respectively, for adolescents and children. Their values are reported in table B2.

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